Chile
Chilean Competition Authority

I. Using a graph database to efficiently ensure compliance

The Compliance Enforcement Division of the Fiscalía Nacional Económica ('FNE') is responsible for the enforcement of mitigation measures established in concentration proceedings, compliance with resolutions of the Competition Tribunal or the Supreme Court, the prosecution of other infringements related to interlocking directorates and gun-jumping, and compliance with the minority shareholding notification obligation, contemplated in Article 4 bis of Decree Law Nr. 211. In particular, the Division has developed substantial experience with computational tools for the detection of infringements to Article 4 bis. The article establishes a legal obligation to inform the FNE of any acquisition of shares or capital in competitors that meets the following requirements:

- The acquisition exceeds 10% of the shares or capital of the competing company.
- Both the acquirer and the acquiree, or indistinctly each business group, have exceeded 100,000 UTM (or USD 7MM app.) in revenues during the last calendar year.

The enforcement of Article 4 bis has been done through a platform specially created to detect minority shareholdings that fall above the preceding thresholds. This platform uses information from several state bodies such as the Chilean Income Tax Service and the Financial Market Commission. The data gathered is processed using *Neo4j*, an open-source graph database.[13]

Neo4j allows us to build conglomerate structure charts that show economic agents linked by their ownership relations. The application can perform specific and general queries, such as the display of the nearest neighbors, identifying a particular identification number of a company, or filtering all the minority shareholdings that have a mandatory obligation under Article 4 bis. To illustrate the potential of this tool, *Figure I*—top—shows the corporate group of an economic agent present in several sectors of the economy, while the bottom displays a zoom-in of the nodes.

[13] Neo4j, *Neo Technology*, https://neo4j.com/ (last accessed June 7, 2023).

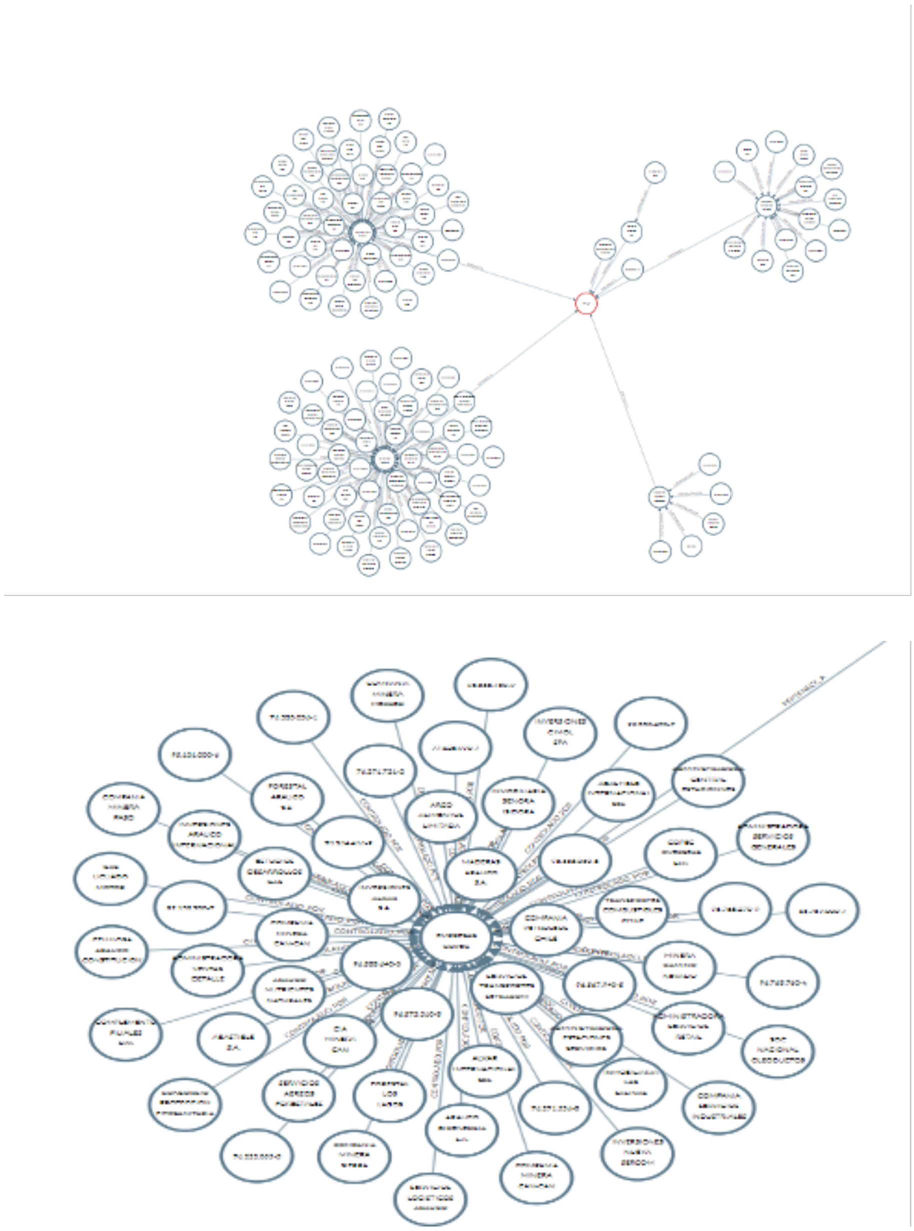

Figure I: Visualization of ownership relations.
Top: Corporate group.
Bottom: Zoom-in.

In this representation, which technically corresponds to a *directed graph*, each node corresponds to a natural or juridical person while the link between two nodes represents a shareholder relation. The direction of the arrow indicates that the agent at the *tail* (or origin) owns shares of the agent at the *head* (or destination). When clicking over a node, a list of information is displayed, including the economic activity of the respective person and its annual sales.

The tool allows us to filter all shareholdings between competing entities that met the conditions stated above for Article 4 bis and were classified under the same economic sector obtaining a list of potential cases to analyze. This procedure is straightforward and does not require much coding, as can be seen in the code snippet in *Figure II*. The result is a time saving application that contributes to a more efficient detection, considering that the alternative would have been a time and resource-consuming approach based on case-by-case examinations, or on eventual formal denouncements.

Notwithstanding, the list must be further refined by human analysis because of potential errors in the database and because economic activities listed in the database are too broad. Therefore, the economic activities must be further specified using mainly public information, to determine that there are in fact rival products or services between the groups linked by the shareholding and, therefore, an infringement.

```
MATCH (g0:Grupo)<-[:PERTENECE_A]-(gc0:Controlador)<-[:CONTROLADO_POR]-(c0:Contribuyente)-
[r:SOCIO_DE]->(c1:Contribuyente)-[:CONTROLADO_POR]->(gc1:Controlador)-[:PERTENECE_A]->(g1:Grupo),
        (g0)<-[:ACTIVIDAD_DE]-(a:Actividad)-[:ACTIVIDAD_DE]->(c1)
WHERE g0.venta_min_UF > 100000
AND c1.venta_min_UF > 100000
AND r.participacion >= 10
AND g0 <> g1
WITH g0, gc0, c0, r, c1, gc1, g1, a
MATCH (a)-[:ACTIVIDAD_DE]->(d:Contribuyente)-[:CONTROLADO_POR]->()-[:PERTENECE_A]->(g0)
RETURN g0.nombre as `Grupo declarante`,
c0 as declarante,
r.participacion as Participacion,
c1 as adquirido,
g1.nombre as `Grupo adquirido`,
a.act_econ as `Act. econ.`,
d as competidor
```

Figure II: Shareholdings filtering query example

In 2020, after a month of implementation, the FNE filed the first two lawsuits supported by this tool against Inversiones Los Orientales[14] and Banmedica,[15] for the minority interest they acquired in competitors without informing the FNE. In both cases the FNE agreed to an extrajudicial settlement where the companies involved committed to pay a substantial fine.

[14] FNE, *Lawsuit against Inversiones Los Orientales*, https://www.fne.gob.cl/wp-content/uploads/2020/01/Requerimiento-FNE-Inv-Los-Orientales.pdf (last accessed June 7, 2023).

[15] FNE, *Lawsuit against Banmédica*, https://www.fne.gob.cl/wp-content/uploads/2020/01/Requerimiento-FNE-Banmedica.pdf (last accessed June 7, 2023).

Still, there are some challenges ahead such as getting more precise data of the economic activities carried out by companies and their annual sales and accessing up-to-date data on the ownership relations each time a query is made. Another medium-term objective is to extend the capabilities of the application to detect companies engaging in gun-jumping behavior.

II. Creation of the Intelligence Unit

The Intelligence Unit ('IU') of the Fiscalía Nacional Económica was launched in September 2020, and has a staff of three professionals, including two data scientists. Through technological tools and analytical techniques, the IU focuses on cartel screening and the development of investigative approaches for the analysis of large volumes of data, structured and unstructured.

The FNE has prioritized an in-house development strategy to embed, in these tools and techniques, the specific knowledge of lawyers and economists working in the office, and, by doing so, fulfilling in a better manner the investigative needs of the institution.

Below we present the IU's main ongoing projects.

A. Cartel detection

- Automated web-based data collection tools to collect information that could help answer consumer complaints, improve the data used in ongoing investigations, and build ex-officio investigations. We are currently developing the implementation of *Machine Learning* algorithms and other *Artificial Intelligence* approaches to help us analyze and classify the processed information with a focus on detection.
- Development of a platform to analyze public tenders based on screening techniques, which requires sophisticated data collection and processing algorithms to display simple and useful reports in a user-friendly interface, reducing the time required to gather and review large quantities of data and identify anticompetitive behavior in this kind of procurement mechanism. In a future stage of this project, Machine Learning algorithms could be implemented to build classification models that use the available historical data to help assess the existence of bid-rigging practices in public tenders and build ex-officio investigations.

B. Investigative systems for data analysis

- Systematization of data provided by public services, such as the Customs National Service. In a future stage of this project, this data should be uploaded on a platform that can make it accessible for all the divisions of the FNE to use it as they see fit to perform the analysis required to accomplish their tasks. Public and private alliances are key to develop search engines that give the staff useful data for their investigations, reducing considerably the timed needed to find relevant information.
- Systematization and analysis of phone calls logs and georeferenced data obtained from mobile devices, using programming tools to manage large datasets and create valuable reports to help ongoing investigations by identifying useful trends to improve the decision-making processes and the storytelling of the cases.